25FEBS101IAEII25

Seat No. : .....

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, November 2025

ASH - SEMESTER I

25FEBS101 - Engineering Mathematics – I

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)State expansion of $(1+x)^m$ around $x=0$.	1	CO4	L1
	b)$\Gamma(\frac{7}{2})$ =? $\frac{15}{8}\sqrt{\pi}$ ✗	1	CO5	L1
	c)What happens if $f(x_n)=0$ in the Newton-Raphson method?	1	CO6	L1
	d)$lim_{x\to 0}(\frac{tan^{-1}x}{x})$ = 0 ✗	1	CO4	L1
	e)$\Gamma(n+1)$ =?	1	CO5	L1
	f)What happens if $f(x_0)$ or $f(x_1)$ is zero in the Regula Falsi method?	1	CO6	L1
Q2	a)**Apply** the **Maclaurin's series** to **show that** $sec^2x = 1+x^2+\frac{2x^4}{3}+\ldots$	5	CO4	L3
	OR			
	b)**Analyze** the given **indeterminate form** to identify the suitable method for evaluation, and then **evaluate** $lim_{x\to 1}\ log(1-x).cot\left(\frac{\pi x}{2}\right)$ 0	5	CO4	L4
Q3	a)**Apply** the properties of the **Gamma function** to **evaluate** $\int_0^1 (log\ x)^4\ dx$	5	CO5	L3
	24 1024 OR			
	b)**Apply** the definition and properties of the **Beta function** to **prove that** $\int_0^{\pi} \sin^2\theta(1+\cos\theta)^3\ d\theta = \frac{7\pi}{8}$	5	CO5	L4
Q4	a)**Apply** the **False Position (Regula Falsi) method** to find the **approximate root** of the given equation $x^3-x-1=0$, performing **three iterations.**	5	CO6	L3
	OR			
	b)**Apply** the **Newton–Raphson method** to find the **root of the given equation** $3x-\cos x-1=0$, correct up to **three decimal places.**	5	CO6	L3